NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH POUND

MIL-PRF-83504/4D
27 OCTOBER 1993
SUPERSEDING
MIL-S-83504/4C
26 January 1990

PERFORMANCE SPECIFICATION SHEET

# SWITCH, SLIDE, DUAL IN-LINE PACKAGE (DIP), SINGLE AND DOUBLE POLE, SINGLE THROW STATIONS, SEALED BASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-83504.

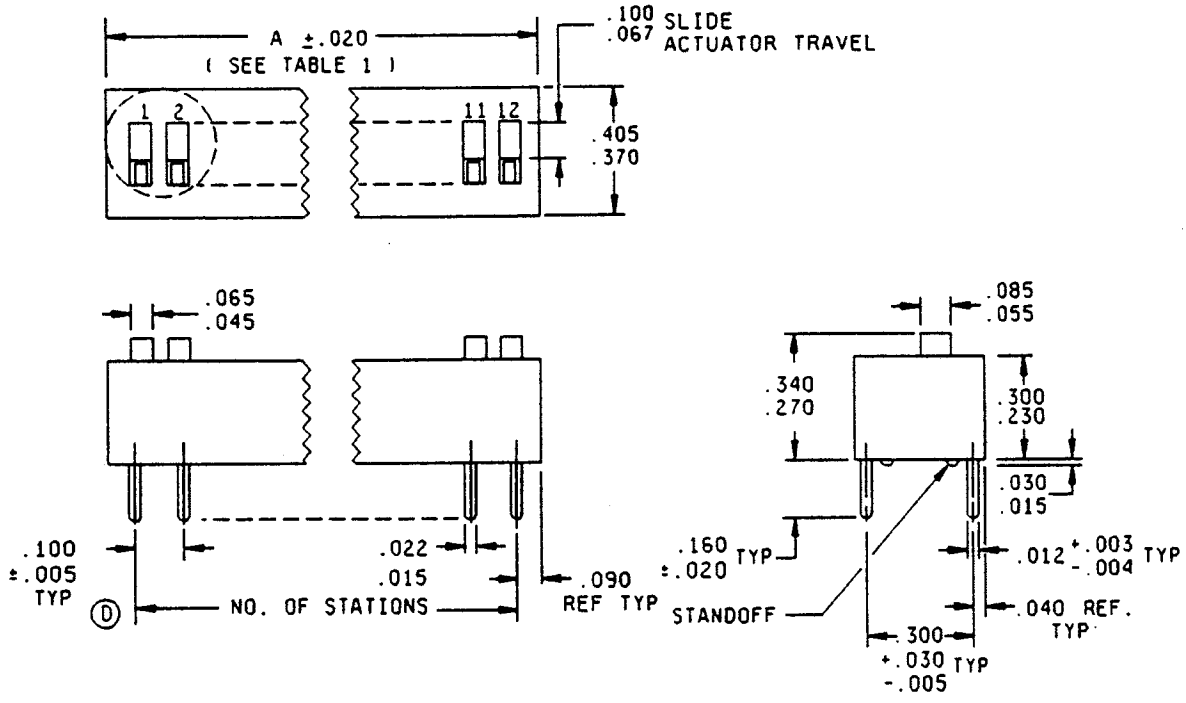
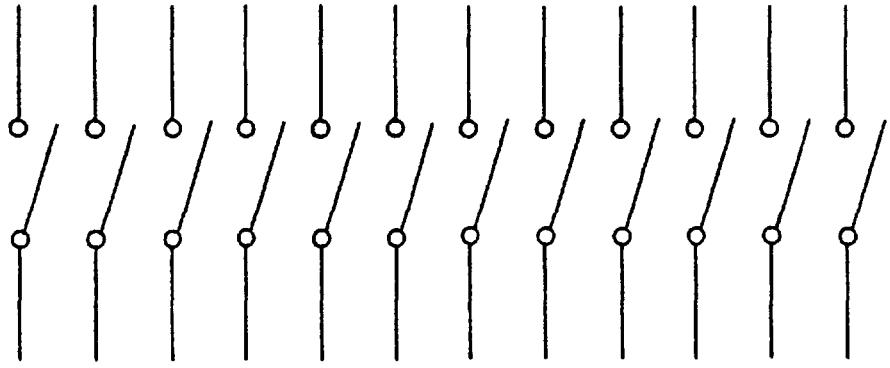



FIGURE 1. Dimensions and configuration, single pole stations (SPST).

(D) denotes changes.


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TYPICAL CONTACT CIRCUIT ARRANGEMENT 12 STATION SWITCH
( CONTACTS SHOWN IN OPEN POSITION )

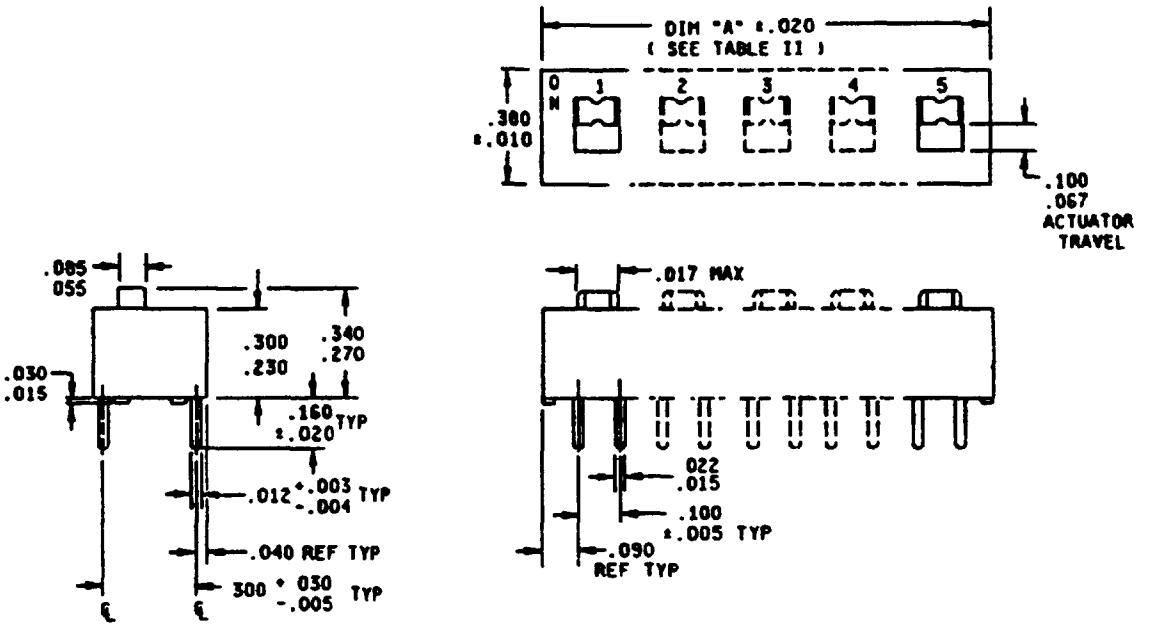
| Inches | mm | Inches | mm |
|---|---|---|---|
| .003 | 0.08 | .067 | 1.70 |
| .004 | 0.10 | .085 | 2.16 |
| .005 | 0.13 | .090 | 2.29 |
| .012 | 0.30 | .100 | 2.54 |
| .015 | 0.38 | .160 | 4.60 |
| .020 | 0.51 | .230 | 5.84 |
| .022 | 0.56 | .270 | 6.86 |
| .030 | 0.76 | .300 | 7.62 |
| .040 | 1.02 | .340 | 8.64 |
| .045 | 1.14 | .370 | 9.40 |
| .055 | 1.40 | .405 | 10.29 |
| .065 | 1.65 | | |

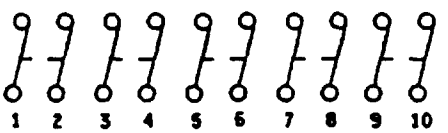
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Each switch station is shown in the "OFF" position. Continuity is established (switch is turned "ON") when the slide actuator toward the station identification number.
4. Exact shape of switch is optional provided outline dimensions are not exceeded.
5. Each switch station shall be numbered as shown.
6. Design of terminals and standoffs are optional.

FIGURE 1. Dimensions and configuration, single pole stations (SPST) - Continued.



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CIRCUIT DIAGRAM - 5 STATION SWITCH
AS VIEWED IN THE POSITION SHOWN
( CLOSED POSITION )

| Inches | mm | Inches | mm |
|---|---|---|---|
| .003 | 0.08 | .065 | 1.65 |
| .004 | 0.10 | .067 | 1.70 |
| .005 | 0.13 | .085 | 2.16 |
| .010 | 0.25 | .090 | 2.29 |
| .012 | 0.30 | .100 | 2.54 |
| .015 | 0.38 | .160 | 4.60 |
| .017 | 0.43 | .230 | 5.84 |
| .020 | 0.51 | .270 | 6.86 |
| .022 | 0.56 | .300 | 7.62 |
| .030 | 0.76 | .340 | 8.64 |
| .040 | 1.02 | .380 | 9.65 |
| .055 | 1.40 | .405 | 10.29 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Each switch station is shown in the "ON" position. Continuity is broken (switch is turned "OFF") when the slide actuator is away from the station identification number.
4. Exact shape of switch is optional provided outline dimensions are not exceeded.
5. Each switch station shall be numbered as shown.
6. Design of terminals and standoffs are optional.

FIGURE 2. Dimensions and configuration, double pole stations (DPST).



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REQUIREMENTS

Material

Body and slide actuator. Molded plastic

(D) Terminals: Gold plated or solderable. See tables I and II The solderable terminals shall be solder over nickel over copper over brass Both gold plated and solderable must meet solderability requirements.

(D) Terminal sealing and solderability The terminal potting material meniscus may extend to .030 inch (0.76 mm) maximum beyond the solder standoff

Dimensions and configuration See figures 1 and 2 and tables I and II.

(D) Seal Sealed base. Temporary tape top seal.

Switch position and setting identification: See figures 1 and 2

Orientation See figures 1 and 2

Operating force 2.5 to 32 ounces

Electrical ratings

10 mA maximum at 50 mV, switching (resistive load).

100 mA maximum at 6 V dc, switching (resistive load)

25 mA maximum at 24 V dc, switching (resistive load).

Application note: In applications where the switch will mate with sockets or with panel boards, the user should consider the compatibility of the mating terminals with respect to finished and mechanical configurations

Part or Identifying Number (PIN): M83504/04- (and dash number from table I or table II)

(D) TABLE I. Dash numbers and characteristics, single pole stations (see figure 1)

| Dash number | | Number of stations | Dimension "A" inches (mm) (see figure 1) |
|---|---|---|---|
| Terminal finish | | | |
| Gold 1/ | Solderable | | |
| 001 | 021 | 1 | .280 (7.11) |
| 002 | 022 | 2 | .280 (7.11) |
| 003 | 023 | 3 | .380 (9.65) |
| 004 | 024 | 4 | .480 (12 19) |
| 005 | 025 | 5 | .580 (14 73) |
| 006 | 026 | 6 | .680 (17.27) |
| 007 | 027 | 7 | .780 (19.81) |
| 008 | 028 | 8 | .880 (22.35) |
| 009 | 029 | 9 | .980 (24.89) |
| 010 | 030 | 10 | 1.080 (27.43) |
| 011 | 031 | 11 | 1.180 (29.97) |
| 012 | 032 | 12 | 1.280 (32 51) |

1/ PIN's for switches equipped with gold terminations are inactive for new design



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(D) TABLE II. Dash numbers and characteristics, double pole stations (see figure 2).

| Dash number solderable | Number of stations | Dimension "A" inches (mm) (see figure 2) |
|---|---|---|
| 041 | 1 | .280 (7.11) |
| 042 | 2 | .480 (12.19) |
| 043 | 3 | .680 (17.27) |
| 044 | 4 | .880 (22.35) |
| 045 | 5 | 1.080 (27.43) |

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AV
Navy - AS, SH
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5930-1503-03)



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